Supplementary Material — circulation architectures across planetary bodies

S1. Methods detail

Complete 56-cell evidence table: planetary_circulation_master.csv; measurement dictionary: measurement_dictionary.csv; observability classes and measurement tiers defined in Tables S1–S2. Status vocabulary and coding conventions follow the study protocol (STUDY_PROTOCOL.md in the repository).

S2. Master evidence table

body	edge_id	process_name	current_activity_status	measurement_tier	observability_class	current_activity_estimate	current_activity_unit	last_major_activity_age_Gyr	primary_reference
Earth	E1	Interior->Surface	direct_active	B	O3	26.5	km3/yr silicate magma emplaced+erupted		Crisp 1984 J. Volcanol. Geotherm. Res. 20:177 (subaerial+submarine magma rates); measured eruptions
Earth	E2	Crust/Ice deformation & recycling	direct_active	A	O3	5.8	cm/yr mean relative plate motion		NUVEL-1 plate model (DeMets et al. 1990

									GJI 101:425); GNSS
Earth	E3	Surface<->Atmosphere	direct_active	A	O3	496000.0	km3/yr evaporation=global hydrologic turnover		Trenberth et al. 2007 J. Hydrometeorol. 8:758
Earth	E4	Subsurface Liquid<->Crust/Ice/Surface	direct_active	B	O3		hydrothermal+groundwater exchange		seafloor hydrothermal flux ~10^13 W advective (Elderfield & Schultz 1996 Annu. Rev. Earth Planet. Sci. 24:191)
Earth	E5	Atmosphere->Surface deposition	direct_active	B	O3	17000000000.0	kg/yr aeolian dust emission/deposition		global dust cycle ~1000-3000 Tg/yr (Shao et al. 2011 Aeolian Res. 3:181)
Earth	E6	Planet->Space	direct_active	B	O3	3.0	kg/s H escape (domi		polar wind + Jeans escap

							nant; He ~0.05 kg/s)		e measurements (e.g. Slapak et al. 2017; Catling & Kasting 2017)
Earth	E7	Chemical/ redox recycling	direct_active	A	O3		oxygenic photosynthesis-driven redox cycle		global C cycle inventories (IPCC AR6); redox state measurable directly
Venus	E1	Interior->Surface	probable_active	C	O2		volcanic resurfacing (contemporary)	0.001	Herrick & Hensley 2023 Science 379:1205 (Maat Monsvent change, 1991); transient SO2 (Marcq et al. 2013 Nat. Geosci. 6:25);

									mean surface age 0.15-0.75 Ga (Phillips et al. 1992 JGR 97:15923)
Venus	E2	Crust/Ice deformation & recycling	probable_active	C	O2		rifting / tesserae deformation	0.01	geologically recent rifting (e.g., Parga Chasma); no direct measurement
Venus	E3	Surface<->Atmosphere	residual	C	O2		chemical weathering; possible volcanic degassing		surface-atmosphere reactions; SO2 variability as degassing proxy
Venus	E4	Subsurface Liquid<->Crust/Ice/Surface	structurally_absent	D	O0				no subsurface liquid water reservoir (surface ~740 K)
Venus	E5	Atmosphere-	probable_active	C	O2		H2SO4 aeros		sulfuric acid

		>Surface deposition					ol deposition		cloud cycle (Venus Express VIRTIS/SPICAV)
Venus	E6	Planet->Space	direct_active	B	O3	1e+25	ions/s total ion escape (O+, H+)		ASPERA-4/ Venus Express escape rates ~10^24-10^25 s^-1 (Persson et al. 2018/ 2020)
Venus	E7	Chemical/ redox recycling	probable_active	C	O2		SO2-H2SO4 photochemical cycle; CO oxidation		photochemical models constrained by SO2 profiles
Mars	E1	Interior->Surface	probable_active	C	O2		volcanism (possible present-day magma)	0.01	Cerberus Fossae eruptions <~10 Ma (e.g., Vaucher et al. 2009 Icarus 204:418);

									InSight suggests subsurface magma (e.g., 2023-24 analyses of LF quakes)
Mars	E2	Crust/Ice deformation & recycling	direct_active	A	O3		marsquake seismicity (~1300 events catalogued)		InSight SEIS Marsquake Service (Clinton et al. 2021 PEPI 310:106595; Banerdt et al. 2020 Nat. Geosci. 13:183)
Mars	E3	Surface<->Atmosphere	direct_active	A	O3		seasonal CO2 cycle (~25-30% atmosphere exchanged)		Viking/Phoenix pressure cycles; GRS seasonal caps; H2O residual cycle

Mars	E4	Subsurface Liquid<->Crust/Ice/Surface	probable_active	C	O2		deep groundwater; ice-table exchange		InSight seismic inference of liquid water ~11-20 km (Wright et al. 2024 PNAS 121:e2409983121); polar basal liquid claims disputed
Mars	E5	Atmosphere->Surface deposition	direct_active	B	O3		aeolian dust deposition (global storms)		orbital dust devil/track monitoring; atmospheric dust loading measured
Mars	E6	Planet->Space	direct_active	A	O3		escape: H ~1-2 kg/s avg; O sputtering		MAVEN escape rates (Jakosky et al. 2018 Icarus 315:146; ~2x10^24 O+/s sputte

									red)
Mars	E7	Chemical/ redox recycling	residual	C	O2		photochemistry, perchlorate/ oxidant cycling		photochemical models + O2 seasonal cycle (SAM); perchlorate at surface
Moon	E1	Interior->Surface	none_detected	A	O3	0.0	current volcanism: none detected	1.9	Chang'e-5 basalts 2.03 Ga (Li et al. 2021 Nature 600:54); youngest IMPs claimed <100 Ma (Braden et al. 2014 Nat. Geosci. 7:787, debated)
Moon	E2	Crust/ Ice deformation & recycling	residual	B	O3		shallow moonquakes + lobate-scarp thrust faulting	0.05	Apollo seismic network (deep+shallow moonquake

									s); young thrust faults <~50 -500 Ma (Watters et al. 2019 Nat. Geosci. 12:411)
Moon	E3	Surface<->Atmosphere	residual	C	O2		exospheric migration of H2O/OH; polar ice sublimation		surface-boundary exosphere; diurnal OH/H2O (LRO/LAMP, M3)
Moon	E4	Subsurface Liquid<->Crust/Ice/Surface	structurally_absent	D	O0				
Moon	E5	Atmosphere->Surface deposition	none_detected	C	O1		no atmosphere->surface deposition (space weathering instead)		
Moon	E6	Planet->Space	direct_active	B	O3		sputtering + exospheric		LADEE exosphere

							escape (continuous)		(Na, K, He, Ar); solar-wind sputtering
Moon	E7	Chemical/ redox recycling	residual	C	O1		space weathering reduction (nanophase Fe)		optical maturity studies
Mercury	E1	Interior->Surface	none_detected	B	O2	0.0	current volcanism: none detected	3.5	major effusive volcanism ended ~3.5 Ga (Byrne et al. 2016 GRL 43:7408); small pyroclastic vents possibly younger
Mercury	E2	Crust/ Ice deformation & recycling	probable_active	B	O3		ongoing global contraction (small graben, scarp slip)	0.0	MESSENGER imagery: <~300 Ma graben => present-day tectonism (Watters et al. 2016

									Nat. Geosci. 9:743)
Mercury	E3	Surface<->Atmosphere	residual	C	O2		hollow volatile sublimation; polar ice		hollows = ongoing volatile loss (Blewett et al. 2013 Science 341:1082); polar ice (Lawrence et al. 2013 Science 339:292)
Mercury	E4	SubsurfaceLiquid<->Crust/Ice/Surface	structurally_absent	D	O0				
Mercury	E5	Atmosphere->Surface deposition	residual	C	O2		exospheric Na/K deposition-recycling		Na exosphere source/sink balance (ground-based + MASCS)
Mercury	E6	Planet->Space	direct_active	B	O3		sputtering + magnetospheric		MESSENGER FIPS/MASC

							cusp loss		S exosphere; magnetospheric recycling
Mercury	E7	Chemical/ redox recycling	residual	C	O1		space weathering; hollow volatile cycling		
Io	E1	Interior->Surface	direct_active	A	O3	9300000000000.0	W global volcanic heat flow		Veeder et al. 2012 Icarus 219:701 (mean ~2.24 W/m2); Davies 2007 'Volcanism on Io'; asthenosphere partial-melt layer (Khurana et al. 2011 Science 332:1186) counted under E1, not E4
Io	E2	Crust/	direct	B	O3		resurf		plume

		Ice deformation & recycling	_active				acing rate ~1 cm/yr equivalent		+PPR deposits; mountain faulting (Jaeger et al. 2003 JGR 108:5093); surface recycling ~cm/yr (Johnson et al. 1979 Science 206:927)
Io	E3	Surface<->Atmosphere	direct_active	B	O3		SO2 frost sublimation/condensation cycle (diurnal collapse)		SO2 atmosphere diurnal collapse measured (de Pater et al. 2020 Planet. Sci. J. 1:29); plume-fed
Io	E4	Subsurface Liquid<->Crust/Ice/Surface	structurally_absent	D	O0		no persistent subsurface volatile/fluid reserv		no water/brine/methane ocean on Io; silicat

							oir		e partial melt is E1 interior-state evidence, not a volatile reservoir
Io	E5	Atmosphere->Surface deposition	direct_active	B	O3		plume fallout + frost deposition		deposits mapped around vents (Galileo SSI/Voyager)
Io	E6	Planet->Space	direct_active	B	O3	1000.0	kg/s material supply to Io plasma torus		torus supply ~1 t/s (Thomas et al. 2004 'Io: the plasma torus' review; Bagenal & Dols 2020)
Io	E7	Chemical/redox recycling	residual	C	O1		sulfur allotrope/redox cycling, photolysis		SO2 photochemistry models
Europa	E1	Interior-	probable_ac	C	O1		possible	0.001	Roth et al.

		>Surface	tive				plumes/cryovolcanic venting (contested)		2014 Science 343:171 (HST UV plumes, unrepeated); thermal anomaly at Pwyll (Trumbo et al. 2017 Sci. Rep. 7:10686); JUICE/Europa Clipper pending
Europa	E2	Crust/Ice deformation & recycling	probable_active	C	O2		ice-shell tectonics: bands, chaos, diapirism	0.03	surface age ~40-90 Myr (Bierhaus et al. 2009 in 'Europa' book); chaos/bands require ongoing renewal
Europ	E3	Surfa	proba	C	O2		sputte		O2

a		ce<->Atmosphere	ble_active				r-produced O2 exosphere; surface-atmosphere exchange		atmosphere (Hall et al. 1995 Nature 373:677); HST coronal imaging
Europa	E4	Subsurface Liquid<->Crust/Ice/Surface	probable_active	C	O2		ocean-ice shell exchange (brine transport, chaos melt-through)		induction -> subsurface ocean (Kivelson et al. 2000 Science 289:1340); chaos models (Sotin et al. 2002 GRL 29:74); Schmidt et al. 2011 Nature 479:502
Europa	E5	Atmosphere->Surface deposition	probable_active	C	O2		radiolytic product deposition (H2SO4 hydrate etc.)		Carlson et al. 1999 Science 286:97; NIMS/Galileo

									mapping
Europa	E6	Planet->Space	probable_active	C	O2		sputtering/ escape of O2 atmosphere		sputtering source of exosphere; escape rates modelled
Europa	E7	Chemical/ redox recycling	probable_active	C	O1		radiolytic oxidant cycle (surface->ocean delivery modelled)		Hand et al. 2007 Astrobiology 7:1006 oxidant budget
Titan	E1	Interior->Surface	probable_active	C	O1		possible cryovolcanism (Doom Mons; disputed)	0.05	Sotin et al. 2005 Nature 435:786; Lopes et al. 2020 Nat. Astron. 4:228 (Sotra/Doom Mons); no observed eruption
Titan	E2	Crust/ Ice deformation & recycl	probable_active	C	O1		tectonic features (graben,		Radar mapping (Lopes et al.

		ing					ridges) modest evidence		2020)
Titan	E3	Surface<->Atmosphere	direct_active	B	O3		CH4 hydrological cycle: lakes, rain, evaporation		Cassini: hydrocarbon lakes/seas (Stofan et al. 2007 Nature 445:61); rain events (Turtle et al. 2011 Science 331:1414); Ontario Lacus
Titan	E4	Subsurface Liquid<->Crust/Ice/Surface	probable_active	C	O2		subsurface water ocean (Lovek2~0.6) - shell exchange		Iess et al. 2012 Science 337:457; Mitri et al. 2014 Icarus 236:169
Titan	E5	Atmosphere->Surface deposition	direct_active	B	O3		organic haze/aerosol deposition		haze production ~10^14 kg/Myr-

							(dune-building flux)		order (Waite? see Tomasko et al. 2005); dune fields global (Lorenz et al. 2006 Science 312:724)
Titan	E6	Planet->Space	direct_active	B	O3		H2 escape dominant; CH4/H ion escape measured		Cassini INMS escape fluxes (Cui et al. 2009/2012); ~10^28 amu/s order
Titan	E7	Chemical/redox recycling	direct_active	B	O3		N2-CH4 photochemistry -> complex organics/haze		Cassini INMS/CIRS photochemical products (Horst 2017 JGRE 122:432)

S3. Sensitivity analyses

Leave-one-body-out PC explained-variance fractions:

excluded	PC1	PC2	PC3

Earth	0.580	0.225	0.130
Venus	0.579	0.174	0.157
Mars	0.592	0.167	0.120
Moon	0.465	0.226	0.175
Mercury	0.529	0.188	0.165
Io	0.588	0.173	0.117
Europa	0.663	0.191	0.090
Titan	0.595	0.180	0.166

S4. Monte Carlo persistence sensitivity

Primary statistic (same as main text): within-body persistence spread = max − min persistence fraction over dated cessations and right-censored cells (censored cells = 1.0). Inferred cessations without a measured date (Venus's dynamo) are excluded. 4000 Monte Carlo draws sample each dated cessation's fraction uniformly within its geological interval; the table gives the median and 5–95% band of the resulting spread.

body	spread_med	spread_lo	spread_hi
Earth	0.000000	0.000000	0.000000
Venus	0.000000	0.000000	0.000000
Mars	0.915582	0.911574	0.919521
Moon	0.495804	0.317008	0.746918
Mercury	0.832638	0.783453	0.883531
Io	0.000000	0.000000	0.000000
Europa	0.000000	0.000000	0.000000
Titan	0.000000	0.000000	0.000000

S5. Reference audit

All cited sources were verified to exist with correct bibliographic details; see literature/reference_audit.csv in the repository.